Annexure

```
//  To derive LST from the LANDSAT 8
image collection
//cloud mask
function maskL8sr(image) {
  // Bits 3 and 5 are cloud shadow and cloud,
respectively.
  var cloudShadowBitMask = (1 << 3);
  var cloudsBitMask = (1 << 5);
  // Get the pixel QA band.
  var qa = image.select('pixel_qa');
  // Both flags should be set to zero,
indicating clear conditions.
  var mask =
qa.bitwiseAnd(cloudShadowBitMask).eq(0)

.and(qa.bitwiseAnd(cloudsBitMask).eq(0));
  return image.updateMask(mask);
}
// Region of interest (ROI)
var ROI= ee.Geometry(geometry);
Map.centerObject(ROI, 11);

// Visualisation to true colour
var vizParams2 = {
bands: ['B4', 'B3', 'B2'],
min: 0,
max: 3000,
gamma: 1.4,
};

// Landsat 8 SR image
 var l8_image= imageCollection
 .filterBounds(geometry)
 .filterDate('2020-01-01','2020-12-31')
 .filterMetadata('CLOUD_COVER',
'less_than', 90)
 .map(maskL8sr)
 .median()
 .clip(ROI);
print(l8_image, 'l8_image');
Map.addLayer(l8_image, vizParams2,
'l8_image');

// NDVI
{
var ndvi =
l8_image.normalizedDifference(['B5',
'B4']).rename('NDVI');

var ndviParams = {min: -1, max: 1,palette: [
   'FFFFFF', 'CE7E45', 'DF923D', 'F1B555',
'FCD163', '99B718', '74A901',
   '66A000', '529400', '3E8601', '207401',
'056201', '004C00', '023B01',
   '012E01', '011D01', '011301'
  ]};

print(ndvi,'NDVI');
}
```

```
// Mean NDVI
{ var mean_NDVI =
ndvi.reduce(ee.Reducer.median())
                .clip(ROI);
Map.addLayer(mean_NDVI,
ndviParams, 'NDVI');
}
{
var meanNDVI =
ee.Number(mean_NDVI.reduceRegion
({
reducer: ee.Reducer.median(),
scale: 30,
reducer: ee.Reducer.median(),

scale: 30,
maxPixels: 1e9
}).values().get(0));
print(meanEm, 'meanEm');
}

/ Min and Max of Emissivity
{
var min_em =
ee.Number(mean_em.reduceRegion({
reducer: ee.Reducer.min(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(min_em, 'min_em');

var max_em =
ee.Number(mean_em.reduceRegion({
reducer: ee.Reducer.max(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(max_em, 'max_em');
}

// Brightness temperature
var brightness_temp=
l8_image.select('B10').multiply(0.1);
var b10Params = {min: 291.918, max:
309.500, palette: ['040274', '040281',
'0502a3', '0502b8', '0502ce', '0502e6',
   '0602ff', '235cb1', '307ef3', '269db1',
'30c8e2', '32d3ef',
   '3be285', '3ff38f', '86e26f', '3ae237',
'b5e22e', 'd6e21f',
   'fff705', 'ffd611', 'ffb613', 'ff8b13',
'ff6e08', 'ff500d',
   'ff0000', 'de0101', 'c21301', 'a71001',
'911003']};
print(brightness_temp,'BT');
```

```
// Mean Brightness temperature
{ var mean_BT =
brightness_temp.reduce(ee.Reducer.median(
))
              .clip(ROI);
Map.addLayer(mean_BT, b10Params, 'BT');
}

// Min and Max of Brightness temperature
{
var min_BT =
ee.Number(mean_BT.reduceRegion({
reducer: ee.Reducer.min(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(min_BT, 'min_BT');

var max_BT =
ee.Number(mean_BT.reduceRegion({
MaxPixels: 1e9
}).values().get(0));
print(meanNDVI, 'meanNDVI');
}
// Min and Max of NDVI
{
var min_ndvi =
ee.Number(mean_NDVI.reduceRegion({
reducer: ee.Reducer.min(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(min_ndvi, 'min_ndvi');

var max_ndvi =
ee.Number(mean_NDVI.reduceRegion({
reducer: ee.Reducer.max(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(max_ndvi, 'max_ndvi');
}
// NDBI
{
var ndbi =
l8_image.normalizedDifference(['B6',
'B5']).rename('NDBI');
var ndbiParams = {min: -1, max: 1, palette:
['F9DFDC', '0A81AB', '0C4271', '000000']};
print(ndbi,'NDBI');
}
// NDBI
{
var ndbi =
l8_image.normalizedDifference(['B6',
'B5']).rename('NDBI');
var ndbiParams = {min: -1, max: 1, palette:
['F9DFDC', '0A81AB', '0C4271', '000000']};
print(ndbi,'NDBI');
}
```

Annexure

```
//  Mean NDBI
{ var mean_NDBI =
ndbi.reduce(ee.Reducer.median())
                .clip(ROI);
Map.addLayer(mean_NDBI, ndbiParams,
'NDBI');
}
{
var meanNDBI =
ee.Number(mean_NDBI.reduceRegion({
reducer: ee.Reducer.median(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(meanNDBI, 'meanNDBI');
}
// Min and Max of NDBI{
var min_ndbi =
ee.Number(mean_NDBI.reduceRegion({
reducer: ee.Reducer.min(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(min_ndbi, 'min_ndbi');
var max_ndbi =
ee.Number(mean_NDBI.reduceRegion({
reducer: ee.Reducer.max(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(max_ndbi, 'max_ndbi');
}
//Fractional vegetation
{
var fv
=(ndvi.subtract(min_ndvi).divide(max_ndvi.sub
tract(min_ndvi))).pow(ee.Number(2)).rename('F
V');
var imageVisParam2 =
{"opacity":1,"palette":["b7efc5","92e6a7","6ede
8a","4ad66d", "2dc653", "25a244", "208b3a",
"1a7431", "155d27", "10451d"]};
print(fv, 'fv');
}
//  Mean Fractional vegetation
{ var mean_fv =
fv.reduce(ee.Reducer.median())
                .clip(ROI);
Map.addLayer(mean_fv, imageVisParam2,
'FV');
}
{
var meanFV =
ee.Number(mean_fv.reduceRegion({
reducer: ee.Reducer.median(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(meanFV, 'meanFV');
}
// Min and Max of Fractional vegetation
{
var min_fv =
ee.Number(mean_fv.reduceRegion({
reducer: ee.Reducer.min(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(min_fv, 'min_fv');

var max_fv =
ee.Number(mean_fv.reduceRegion({
reducer: ee.Reducer.max(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(max_fv, 'max_fv');
}

//Emissivity
var a = ee.Number(0.004);
var b = ee.Number(0.986);
var Emissivity =
fv.multiply(a).add(b).rename('Em');
var imageVisParam3 = {min:
0.9865619146722164,
max:0.989699971371314};
print(Emissivity, 'Em');

//  Mean emissivity
{ var mean_em =
Emissivity.reduce(ee.Reducer.median())
                .clip(ROI);
Map.addLayer(mean_em,
imageVisParam3, 'Emissivity');
}
{
var meanEm =
ee.Number(mean_em.reduceRegion({
reducer: ee.Reducer.median(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(meanEm, 'meanEm');
}

// Min and Max of Emissivity
{
var min_em =
ee.Number(mean_em.reduceRegion({
reducer: ee.Reducer.min(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(min_em, 'min_em');
var max_em =
ee.Number(mean_em.reduceRegion({
reducer: ee.Reducer.max(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(max_em, 'max_em');
}
// Brightness temperature
var brightness_temp=
l8_image.select('B10').multiply(0.1);
var b10Params = {min: 291.918, max:
309.500, palette:
['040274', '040281', '0502a3',
'0502b8', '0502ce', '0502e6',
   '0602ff', '235cb1', '307ef3',
'269db1', '30c8e2', '32d3ef',
   '3be285', '3ff38f', '86e26f', '3ae237',
'b5e22e', 'd6e21f',
   'fff705', 'ffd611', 'ffb613', 'ff8b13',
'ff6e08', 'ff500d',
   'ff0000', 'de0101', 'c21301',
'a71001', '911003']};
print(brightness_temp,'BT');

//  Mean Brightness temperature
{ var mean_BT =
brightness_temp.reduce(ee.Reducer.m
edian())
                .clip(ROI);
Map.addLayer(mean_BT, b10Params,
'BT');
}

// Min and Max of Brightness
temperature
{
var min_BT =
ee.Number(mean_BT.reduceRegion({
reducer: ee.Reducer.min(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(min_BT, 'min_BT');

var max_BT =
ee.Number(mean_BT.reduceRegion({
reducer: ee.Reducer.max(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(max_BT, 'max_BT');
}
```

Annexure

```
//LST (Converted to Celsius from Kelvin)
var LST = brightness_temp.expression(
'(Tb/(1 + (0.00115* (Tb / 1.438))*log(Em)))-273.15', {
'Tb': brightness_temp.select('B10'),
'Em': Emissivity.select('Em')
}).rename('LST');
print(LST, 'LST');

var Visparams_LST = {min: 12.19726940544291,
max:41.6701111498399, palette:
['040274', '040281', '0502a3', '0502b8', '0502ce', '0502e6',
    '0602ff', '235cb1', '307ef3', '269db1', '30c8e2', '32d3ef',
    '3be285', '3ff38f', '86e26f', '3ae237', 'b5e22e', 'd6e21f',
    'fff705', 'ffd611', 'ffb613', 'ff8b13', 'ff6e08', 'ff500d',
    'ff0000', 'de0101', 'c21301', 'a71001', '911003']};

//  Mean LST
{ var mean_LST = LST.reduce(ee.Reducer.median())
                  .clip(ROI);
Map.addLayer(mean_LST, Visparams_LST, 'LST');
}
{
var meanLST = ee.Number(mean_LST.reduceRegion({
reducer: ee.Reducer.median(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(meanLST, 'meanLST');
}
// Min & Max LST
{
var min_LST = ee.Number(mean_LST.reduceRegion({
reducer: ee.Reducer.min(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(min_LST, 'min_LST');

var max_LST = ee.Number(mean_LST.reduceRegion({
reducer: ee.Reducer.max(),
scale: 30,
maxPixels: 1e9
}).values().get(0));
print(max_LST, 'max_LST');
}
// To download Min., Max. & Mean in CSV format
var feature = ee.Feature(null,
{
min_ndvi : min_ndvi, max_ndvi : max_ndvi, meanNDVI : meanNDVI,
min_ndbi : min_ndbi, max_ndbi : max_ndbi, meanNDBI : meanNDBI,
min_ndwi : min_ndwi, max_ndwi : max_ndwi, meanNDWI :
meanNDWI,
min_nmdi : min_nmdi, max_nmdi : max_nmdi, meanNMDI :
meanNMDI,
min_fv : min_fv, max_fv : max_fv, meanFV : meanFV,
min_em : min_em, max_em : max_em, meanEm : meanEm,
min_BT : min_BT, max_BT : max_BT, meanBT : meanBT,
min_LST : min_LST, max_LST : max_LST, meanLST : meanLST
});
// Wrap the Feature in a FeatureCollection for export.
var featureCollection = ee.FeatureCollection([feature]);
// Export the FeatureCollection.
Export.table.toDrive({
  collection: featureCollection,
  description: 'Data',
  fileFormat: 'CSV'
});
// To download image with visualizations.
var thumbnail3 = mean_NDVI.getThumbURL({
  'min':-1,
  'max': 1,
  'palette': [ 'FFFFFF', 'CE7E45', 'DF923D', 'F1B555',
'FCD163', '99B718', '74A901',
    '66A000', '529400', '3E8601', '207401', '056201',
'004C00', '023B01',
    '012E01', '011D01', '011301'],
  'region': ee.Geometry(ROI),
  'dimensions': 500,
  'crs': 'EPSG:3857'
});
print('LST visualization', thumbnail3);

// Export to drive

Export.image.toDrive({
  image: LST,
  description: 'LST',
  folder: "image EE",
  scale: 30,
  region: l8_image,
  fileFormat: 'GeoTIFF',
  formatOptions: {
    cloudOptimized: true
  }
});
```